

Interface Capability Brief

Medical Lab HL7 LLP (ORU → MEDITECH) Demo · Stakeholder overview · 2026-08-05

1. Executive summary

Demo slice of the Large-Scale Laboratory Interoperability case study: lab ORU results inbound over HL7 LLP, validate, then forward to MEDITECH over HL7 LLP with MLLP ACKs.

2. Who this is for

- Sources: Medical Lab (ORU^R01) via MLLP - Destinations: MEDITECH (mock MLLP sink), audit files, SQL log - Demo vs production: Demo only (shared SA password, mock MEDITECH, heuristic/HAPI parse validation — not partner certification)

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: HL7 LLP (HL7TCPLListener) host port 2575
- Format: HL7 v2.x ORU^R01 (pipe-delimited), MLLP framed
- Identity: MSH-10 control ID
- Samples: samples/LAB_ORU_R01_glucose.hl7

Integration highlights

FQCN sources: PilotFish_Documentation tracker (HL7 LLP / HL7 MLLP Simple) + PilotFish_V2 modules.conf / format-hl7 Java. No deep-dive PDF yet for LLP — config from V2 descriptors. Image: pilotfish-eip:23R1 (possible skew vs 26R1 docs).

Stage	Module / mechanism	Notes
Listener	com.pilotfish.eip.modules.hl7.HL7TCPLi stener	Port 2575, Engine=PilotFish, async AutogenerateSynchronousACK=true (immediate AA; sync ACK hung client in smoke test on 23R1)
Archive	FileWriteProcessor	output/inbound/
Validate	com.pilotfish.eip.modules.hl7.HL7Validat ionProcessor	HAPI parse; ThrowException=true (fail-closed)
Audit outbound	FileWriteProcessor	output/meditech-outbound/
Router	com.pilotfish.eip.modules.internal.NullRo utingModule	All Targets — raw HL7 is not XML, so no XPath router
Transport	com.pilotfish.eip.modules.hl7.HL7Simple Transport	Host meditech-mock Port 2576

4. Honest boundaries (not claimed yet)

- None for this slice; future: ORM inbound, ASTM, X12 billing routes.

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Process Lab ORU LLP” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- What is checked: HAPI pipe parse (HL7ValidationProcessor) - What is NOT checked: full lab ORU IG, ASTM, X12 billing from the case study - Failure blocks outbound: yes (ThrowException=true)

6. State & idempotency

- No SQL claim cycle; evidence is files + MLLP ACK - Dedup: demo does not enforce unique ControlId (accepted risk) - Poison: invalid HL7 throws in validation → sync AE ACK; nothing sent to MEDITECH

8. Dual-write / side effects

- Order: archive → validate → outbound file → LLP to MEDITECH - File audit + LLP are sequential processors then transport (not a fan-out dual-write)

3. Inbound contract

- Transport: HL7 LLP (HL7TCPListener) host port 2575 - Format: HL7 v2.x ORU^R01 (pipe-delimited), MLLP framed - Identity: MSH-10 control ID - Samples: samples/LAB_ORU_R01_glucose.hl7

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Process Lab ORU LLP: runs steps such as Lab HL7 LLP Inbound, Archive Inbound ORU, HL7 Parse Validation, Audit Outbound ORU.

7. Security & trust posture

- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Inbound clinical/FHIR payloads can be validated before persistence
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: High — Case study scope reduced — Full case study includes X12 837/835, CSV, positional, ACORD
- Risk noted in design: Med — 23R1 vs V2/26R1 module APIs — Config tags from V2 source
- Risk noted in design: Med — Async AA before MEDITECH completes — Sync ACK did not return to client in smoke
- Risk noted in design: Low — No ControlId uniqueness — Resend duplicates

8. Seeing it work

- - Ports: EIP 8098, Web UI 8099, LLP in 2575, MEDITECH mock 2576 - Volumes: ./output, ./logs, ./samples - Heap: 2GB (no SNIP) - Cold start: ~60–90s EIP
- docker compose up -d --build
- Wait ~60–90s, then open http://localhost:8099/

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- `python3 tools/mlp_send.py --host 127.0.0.1 --port 2575 samples/LAB_ORU_R01_glucose.hl7`
 - `docker compose logs -f meditech-mock pilotfish`

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- LabHL7LLP_V2_Route_Diagrams.pdf — detailed PilotFish route wiring diagrams
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.